

K-FIELD

MASS COUPLING FACTOR

A Foundational Educational Reference on the Complete Scalar Coupling, Mass-Extensive Operator, Cell3 Geometry, Anisotropic Carrier Tensor, Body-Geometry Reductions, Continuous-Rim Closure, and Earth Kinematic Calibration

Cell3 geometry, anisotropic transformation, and calibrated mass coupling

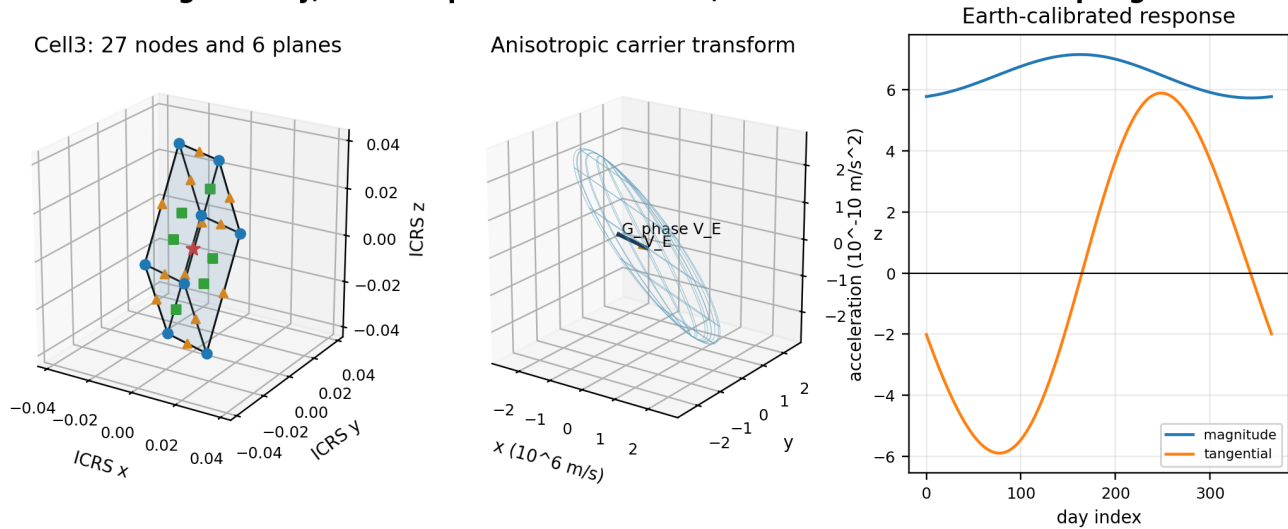


Figure 1. Title composite generated exclusively from the numerical arrays contained in K-Field_derived_properties_1784893730.json.

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Krytronx

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Archival foundational education reference - Unix epoch 1785088666 - 2026-07-26T17:57:46Z

Frontispiece: authoritative base cell, planes, and all nodes

Authoritative centered Cell3 in ICRS native coordinates
Six planes, twelve edges, all 27 half-step nodes, and directional references

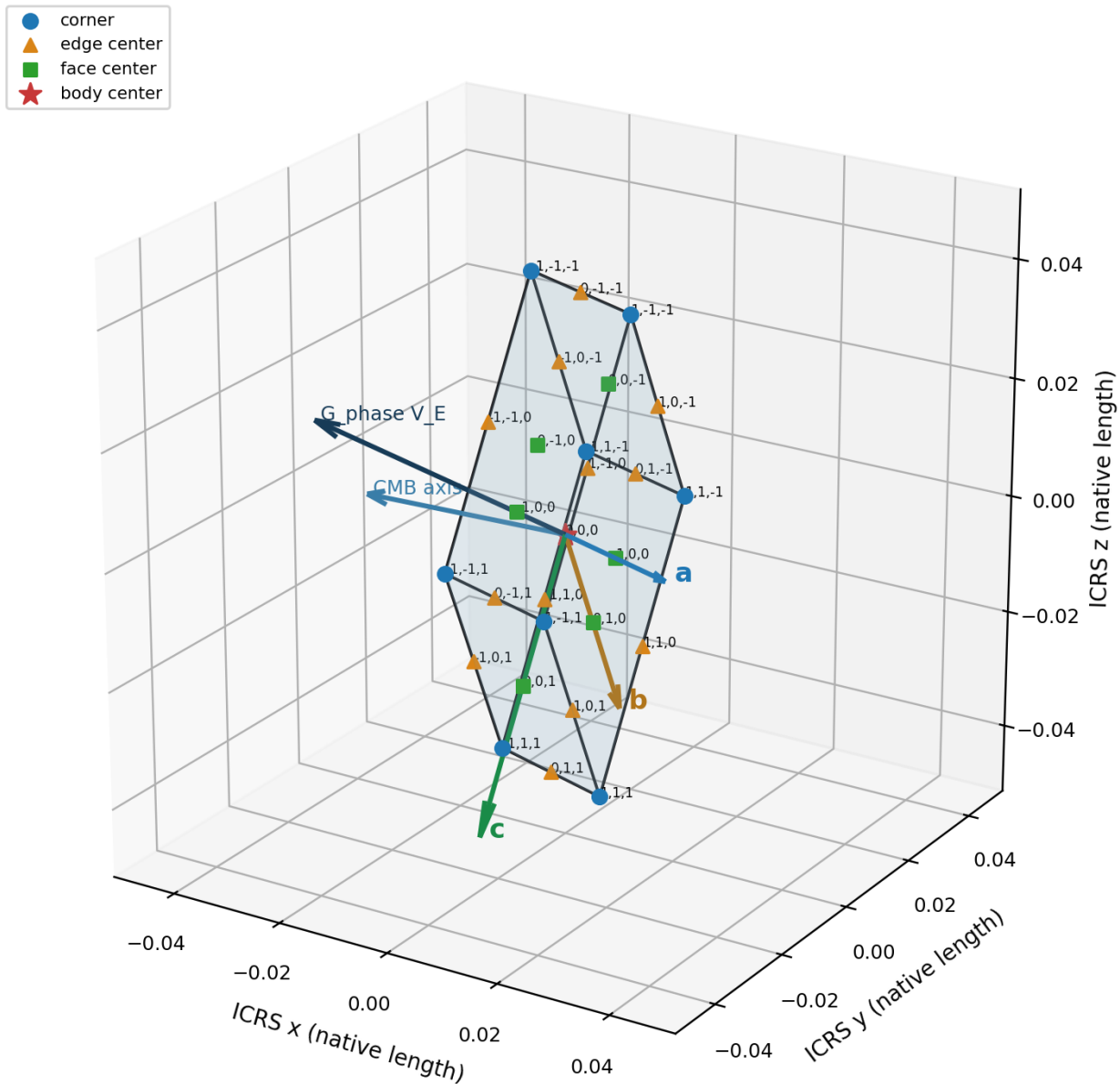


Figure 2. Dimensionally correct centered primitive Cell3 in ICRS native coordinates. All six boundary planes, twelve cell edges, twenty-seven half-step nodes, integer addresses, primitive basis vectors, and scaled directional references are generated from the source arrays.

Boundary-plane atlas

All six boundary planes with complete nine-node half-step arrays

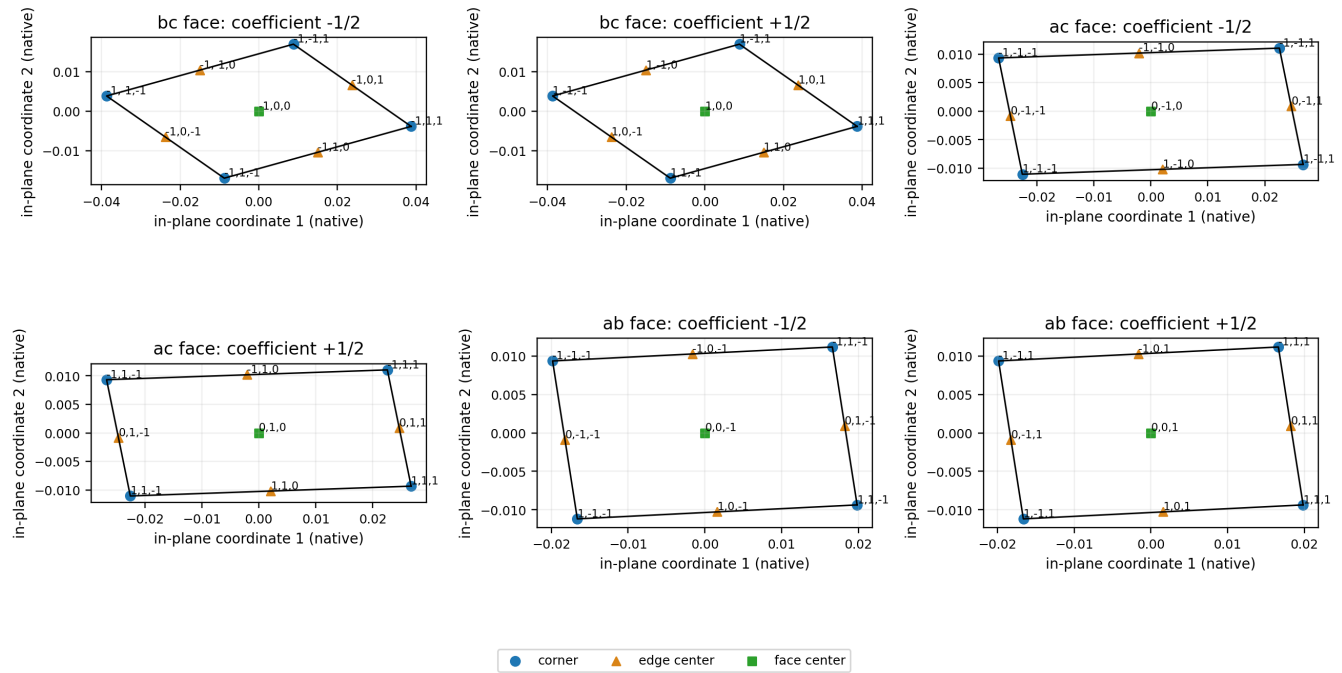


Figure 3. The six boundary planes. Each panel preserves in-plane native distances and contains the complete nine-node half-step array for that face, including exact integer node addresses.

Abstract

This paper isolates and fully develops the K-Field mass-coupling factor contained in the archival source file `K-Field_derived_properties_1784893730.json`. The central scalar is the dimensionless coefficient **$\kappa_F = 3.0740508997020345e-11$** . It is equivalent to **30.740508997020346 parts per trillion** and has inverse **32530365717.006484985352**. The scalar is not the complete coupling by itself. The complete mass coupling is the mass-extensive operator $C_M = M \kappa_F H$, acting on the cross product of angular velocity with the anisotropically transformed translation vector. The whole-body force law is therefore $F_K = M \kappa_F H [\omega \text{ cross}(G_{\text{phase}} V_{\text{CMB}})]$. This decomposition separates the response into a material susceptibility, total coherent mass, body-geometry operator, angular-velocity vector, and carrier-transformed translation state.

The carrier tensor G_{phase} is symmetric and positive definite. Its principal gains are 1.4293287534874834, 2.1125450264474788, and 8.800336434665038, producing a principal gain ratio of 6.156971524705357. It maps the declared Earth-CMB velocity vector of magnitude 369812.50200040644 m/s to a transformed vector of magnitude 930825.2172704348 m/s. The transformed direction is separated from the CMB axis by 31.64086127113854 degrees. The cross-product structure makes the leading response transverse, spin odd, and orientation selective: the response vanishes when ω is parallel to $G_{\text{phase}} V_{\text{CMB}}$ and is maximal when ω is perpendicular to it.

The Earth calibration uses two independent channels. The vector-projection channel gives $3.04273994475465e-11$ and the apsidal-rate channel gives $3.10536185464942e-11$. Their arithmetic midpoint is the recommended coefficient $3.0740508997020345e-11$, with a stored fractional cross-channel spread of 0.020371136307742867. For a spherical body $H=I/3$, producing the scalar prefactor $\kappa_F/3 = 1.0246836332340115e-11$. For an ideal uniform zero-thickness rim, exact antipodal integration gives $F = (\kappa_F M \omega/2) n \text{ cross}(G_{\text{phase}} V_E)$, with scalar prefactor $\kappa_F/2 = 1.5370254498510173e-11$. At maximum orientation the calibrated one-kilogram rim produces $1.430702048307761e-5$ N per rad/s.

Central conclusion

κ_F measures the susceptibility of matter to the locked K-Field kinematic tensor. It is a dimensionless matter-interface coupling. The complete mass coupling is the operator $M \kappa_F H$, not the scalar alone. The source does not identify κ_F as an elastic modulus, an energy density, or an intrinsic field-stiffness measurement.

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1. Purpose, Scope, and Source Integrity

The purpose of this volume is to convert the complete numerical extraction of the K-Field mass-coupling factor into a stand-alone educational reference. Every numerical constant, vector, matrix, plane, node, calibration value, and plotted curve in this report is generated from the source JSON or from algebraic operations explicitly stated in that source. The report does not replace the source data. It exposes the source in a form that can be read, checked, taught, and reproduced.

1.1 Authoritative source record

Field	Value
Source filename	K-Field_derived_properties_1784893730.json
Source SHA-256	140ebb7d755ad09513915aa63404b8e7e3d605b916811fe352fd5d5a84c8258b
Source schema	K-Field.derived_properties.archival_reference.v1
Source epoch	1784893730
Source generation time	2026-07-24T11:48:50Z
Report epoch	1785088666
Report generation time	2026-07-26T17:57:46Z
Principal Investigator	P. Dwayne Esterline
Organization	Krytronx
Classification	Proprietary Confidential Information and Trade Secret of Krytronx, LLC. All Rights Reserved. Do Not Distribute.

The source record contains four linked source-artifact hashes, the full Cell3 geometry, the carrier tensor, the constitutive law, a complete continuous-rim derivation, a 0.1-degree force sweep, a five-degree full-sphere force disk, calibrated engineering quantities, Earth-calibration vectors, and a 366-epoch daily ledger. The report uses these records directly and records the SHA-256 hash of every generated figure in the companion machine-readable file.

1.2 Typed hierarchy of the coupling law

The mass-coupling factor is best understood as a typed hierarchy. A scalar alone cannot specify a vector force. The scalar κ_F first multiplies total coherent mass M and a dimensionless body operator H . That operator acts on a kinematic kernel produced by the cross product of angular velocity with a transformed translation vector. The transformed translation vector is produced by G_{phase} , which carries the anisotropic geometry. The hierarchy is therefore scalar $\rightarrow$ mass and shape operator $\rightarrow$ rotational kernel $\rightarrow$ acceleration $\rightarrow$ force.

$$a_K = \kappa_F H[\omega \text{ cross}(G_{\text{phase}} V_{\text{CMB}})]$$

$$F_K = M a_K = M \kappa_F H[\omega \text{ cross}(G_{\text{phase}} V_{\text{CMB}})]$$

Complete mass-coupling operator

$C_M = M \kappa_F H$. The operator C_M is the complete mass-coupling factor that converts the kinematic vector kernel into force. κ_F alone is the dimensionless material susceptibility contained inside that operator.

2. Mathematical and Dimensional Foundations

2.1 Vectors, tensors, and cross products

A vector carries magnitude and direction. In this work, ω is an angular-velocity vector with units s^{-1} , V_{CMB} is a translation velocity with units m/s , and a_K is an acceleration with units m/s^2 . G_{phase} and H are dimensionless linear operators. Applying G_{phase} to V_{CMB} changes the direction and magnitude of the velocity while preserving its physical units. Taking the cross product with ω then produces m/s^2 . Multiplication by the dimensionless κ_F leaves the units unchanged.

Object	Meaning	Units
κ_F	matter-interface susceptibility	dimensionless
M	total coherent mass	kg
H	body-geometry response operator	dimensionless
ω	angular-velocity vector	s^{-1}
G_{phase}	anisotropic carrier tensor	dimensionless
V_{CMB}	translation velocity	m/s
a_K	whole-body acceleration	m/s^2
F_K	whole-body force	$N = kg \ m/s^2$

2.2 Dimensional audit

$[F_K] = [M][\kappa_F][H][\omega][G_{phase}][V_{CMB}]$
$kg \times 1 \times 1 \times s^{-1} \times 1 \times m \ s^{-1} = kg \ m \ s^{-2} = N$

The dimensional closure is exact. The scalar κ_F must be dimensionless under this constitutive form. The numerical value cannot be compared directly with Newton's gravitational constant G , because G carries units $m^3 \ kg^{-1} \ s^{-2}$ while κ_F does not. The two constants occupy different mathematical roles: G maps two masses and an inverse-square separation into force, whereas κ_F scales a pre-existing rotational-translation kernel into acceleration.

2.3 Transverse rank-two geometry

For a fixed transformed vector $g = G_{phase} V_{CMB}$, the map $\omega \rightarrow \omega \text{ cross } g$ has a one-dimensional nullspace parallel to g and a two-dimensional active subspace perpendicular to g . This is why the source classifies the coupling as transverse rank two. The force is not generally parallel to the CMB velocity, the transformed velocity, or the spin axis. It lies perpendicular to both ω and g before the shape operator H is applied.

$\omega \text{ parallel } g \Rightarrow \omega \text{ cross } g = 0$
$\omega \text{ perpendicular } g \Rightarrow \omega \text{ cross } g = \omega g $

3. Cell3: Geometric Foundation of the Carrier

Cell3 is the oblique three-dimensional base cell used to encode the K-Field geometry. Its primitive basis matrix B has columns a , b , and c . The centered half-step node rule is $p(L) = (1/2) B L$ for integer addresses $L=(i,j,k)$ with i,j,k in $\{-1,0,1\}$. This produces exactly 27 nodes: eight corners, twelve edge centers, six face centers, and one body center. Because the basis is oblique, the cell has six parallelogram boundary planes rather than orthogonal square faces.

3.1 Primitive basis and intrinsic dimensions

Component	a	b	c
ICRS x	0.020381438516946	0.0021987459683234	-0.00063037948534737
ICRS y	-0.0034164144964105	0.012902170258693	-0.024257722017128
ICRS z	-0.002332965916729	-0.034066501536056	-0.042924354571094
Quantity	Value		Native dimension
a	0.0207970587816105		length
b	0.0364942051305875		length
c	0.049308565900292		length
angle(a,b)	83.9334830123133		degrees
angle(a,c)	80.448130018338		degrees
angle(b,c)	50.363231813756		degrees
A_ab	0.000754721789639889		length^2
A_ac	0.00101125575765695		length^2
A_bc	0.00138578443174014		length^2
Cell volume	2.84206693771131e-05		length^3

3.2 Direct and reciprocal metrics

The direct metric $M = B^T B$ records squared edge lengths and pairwise dot products. Its inverse $R = M^{-1}$ is the reciprocal metric. Together they separate the finite address grammar from the physical oblique embedding. The trace of M is 0.004195679333222606, the complete 351-pair squared-distance budget is 0.5097750389865466, and the trace of R is 4348.758045152614. Relative to a rectilinear cell with the same three edge lengths, Cell3 has 0.7594271207104246 times the volume and 1.2517331688608895 times the reciprocal trace.

Cell3 metric structure and carrier principal gains

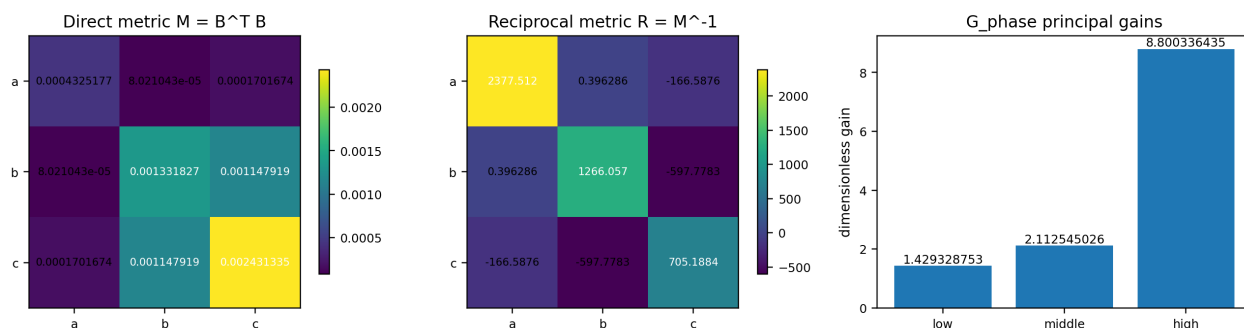


Figure 4. Direct metric, reciprocal metric, and the three principal gains of G_{phase} . Every displayed number is computed from the stored matrices.

3.3 Boundary planes and face normals

Plane family	Unit normal (ICRS)	Positive d	Negative d
ab faces (constant c coefficient)	(0.194092200808, 0.913177168821, 0.358379234791)	-0.018828573500193	0.018828573500193
ac faces (constant b coefficient)	(0.0890525941016, 0.866576769291, -0.491033948326)	0.014052166903391	-0.014052166903391
bc faces (constant a coefficient)	(-0.995965190309, 0.0836021625074, -0.0326192905644)	-0.010254361619054	0.010254361619054

Each boundary plane is represented by $n \cdot x + d = 0$. The two members of each family use the same unit normal and opposite offsets. The exact nine-node arrays on the six faces are shown in Figure 3 and listed in the node appendix. These planes provide the geometric substrate through which body orientation and transformed translation are resolved.

4. The Anisotropic Carrier Tensor G_{phase}

The carrier tensor G_{phase} is the dimensionless linear operator that transforms the declared translation velocity before rotational coupling occurs. It is symmetric, so its eigenvectors form an orthonormal principal-axis basis. All three eigenvalues are positive, so the tensor is positive definite and never reverses the sign of the quadratic form $\mathbf{x}^T G_{\text{phase}} \mathbf{x}$. Its unequal eigenvalues make the carrier strongly anisotropic: equal-speed translation in different directions produces different transformed magnitudes and directions.

row/column	x	y	z
x	2.9764106378	1.2431507795	-2.0803600354
y	1.2431507795	2.5922286635	-2.3178839937
z	-2.0803600354	-2.3178839937	6.7735709133
Principal mode	Eigenvalue	Eigenvector column (ICRS)	
1	1.429328753487483	(-0.408953067284, 0.884525009431, 0.224439070683)	
2	2.112545026447479	(-0.828553895965, -0.256832305478, -0.497529504999)	
3	8.800336434665038	(-0.382434086139, -0.389426083555, 0.837909001746)	

4.1 Translation state and transformed state

Quantity	ICRS vector (m/s)	Magnitude (m/s)
V_E	(-3.5900319906581e+05, 76679.886034002, -44701.060100688)	3.698125020004064e+05
$G_{\text{phase}} V_E$	(-8.8022198163509e+05, -1.4391143656160e+05, 2.6633502694862e+05)	9.308252172704348e+05

The directional gain for the stored Earth velocity is 2.517019333406451. The transformed vector is not collinear with the original CMB direction. The angle between the CMB axis and $G_{\text{phase}} V_E$ is 31.64086127113854 degrees. This directional rotation is essential: the cross product responds to the transformed direction, not to the untransformed velocity alone.

Carrier transformation rotates and amplifies the translation vector

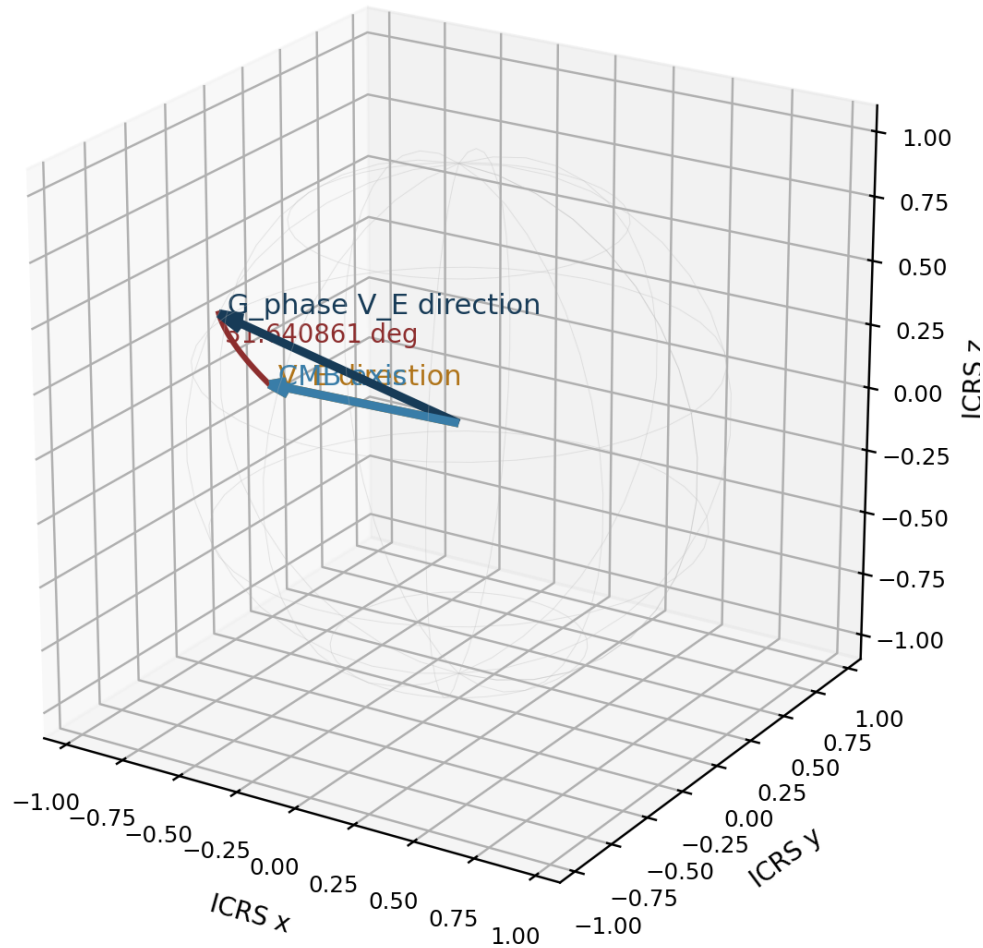


Figure 5. Unit-direction comparison of the declared Earth velocity, the transformed velocity, and the CMB axis. The labeled arc is the exact 31.640861-degree separation between the CMB axis and G_phase V_E.

5. The Complete K-Field Mass-Coupling Factor

The phrase mass-coupling factor can refer to two nested objects. The first is the universal dimensionless scalar κ_F . The second is the complete body-level operator that includes total coherent mass and geometry. The source defines the scalar explicitly, while the force law shows that the complete mass coupling is $M \kappa_F H$. Keeping these two levels separate prevents dimensional and conceptual errors.

5.1 Universal scalar susceptibility

Representation	Exact stored value
Dimensionless coefficient κ_F	3.0740508997020345e-11
Parts per trillion	30.740508997020346
Inverse coupling	32530365717.006484985352
Percent	3.0740508997020347e-09%
One coupled part per	32530365717.006485 parts

$$\kappa_F = 3.0740508997020345e-11 = 30.740508997020346 \text{ parts per trillion}$$

The scalar is small because it represents a suppressed matter response per unit kinematic kernel. The small numerical value does not imply that G_{phase} is weak. The carrier tensor has gains greater than one and a high-to-low ratio of 6.156971524705357. The complete response is the product of a strongly structured carrier transformation and a weak matter-interface susceptibility.

5.2 Complete mass-extensive operator

$$C_M = M \kappa_F H$$

$$F_K = C_M[\omega \text{ cross}(G_{\text{phase}} V_{\text{CMB}})]$$

For a fixed normalized shape, multiplying the coherent mass by a factor r multiplies the force by the same factor r . The acceleration remains unchanged because force and inertial mass scale together. This is the meaning of mass extensivity in the source property map: F is proportional to M , while a_K is independent of M for equal normalized geometry.

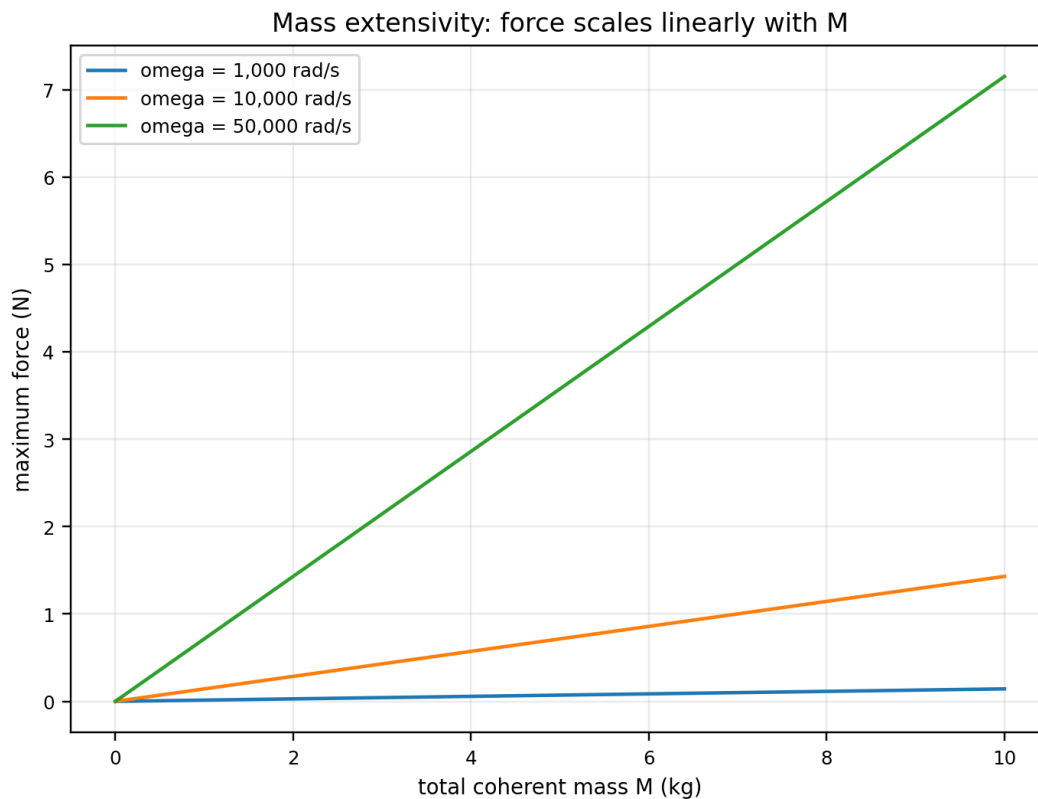


Figure 6. Linear mass extensivity at three angular rates using the calibrated one-kilogram maximum-orientation slope.

5.3 Geometry is part of the coupling

The operator H encodes how the body distributes coherent mass relative to the active transverse plane. It is dimensionless because it is normalized by total mass. A sphere, a rim, a rod, and a general anisotropic body can share the same κ_F while producing different forces because their second moments and coherence geometry differ. Therefore κ_F is universal within the constitutive model, but C_M is body specific.

6. Earth Calibration of kappa_F

The recommended coefficient is obtained from two Earth-only kinematic channels stored in the source. The vector-projection channel compares the direction and magnitude of the K-indexed eccentricity-vector change with the observed one-year change. The apsidal-rate channel compares the unit-response apsidal rate with the observed apsidal rate. Each channel independently solves for the scalar required to map the unit response into the corresponding observed channel.

6.1 Vector-projection channel

$$\text{kappa_vector} = 3.04273994475465\text{e-}11$$

The stored observed one-year eccentricity-vector change is (-8.205803672576249e-7, -5.717914785385164e-7, -2.880126483287629e-7). The unit K-Field response vector per unit kappa is (-29838.8266502998, -5080.1408120101, -2201.1301894097). Projection of the observed change onto the unit-response direction gives the vector-channel coefficient. At the recommended coefficient, the fitted K-indexed vector change is (-9.079178975347679e-7, -1.5457547373681452e-7, -6.697466750922263e-8), leaving the stored residual vector.

6.2 Apsidal-rate channel

$$\text{kappa_apsidal} = 3.10536185464942\text{e-}11$$

The stored observed apsidal rate is 5.6421894029068415e-5 rad/year. The unit-response rate is 1816918.5 rad/year per unit kappa. Their ratio yields the apsidal-channel coefficient. This channel depends on the phase-indexed orbital rotation rather than the direct vector projection, so agreement between the two channels provides an internal cross-channel consistency test.

6.3 Recommended midpoint and interval

$$\text{kappa_F} = (\text{kappa_vector} + \text{kappa_apsidal})/2$$

$$\text{kappa_F} = 3.0740508997020345\text{e-}11$$

Calibration quantity	Value
Vector projection kappa_F	3.0427399447546500e-11
Apsidal-rate kappa_F	3.1053618546494197e-11
Recommended midpoint	3.0740508997020345e-11
Calibration interval	[3.0427399447546500e-11, 3.1053618546494197e-11]
Fractional cross-channel spread	0.02037113630774287
Directional separation	27.47373717145601 deg
K-indexed observed vector power fraction	0.7871635746511932

Two-channel derivation of the recommended mass-coupling coefficient

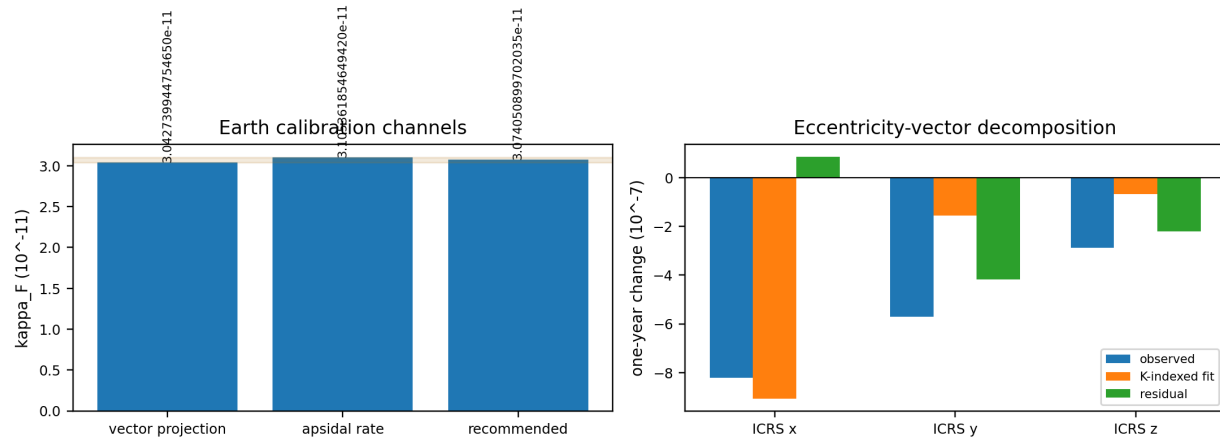


Figure 7. The two Earth calibration channels, the recommended midpoint, and the stored eccentricity-vector decomposition.

7. Spherical-Body Reduction

For a homogeneous isotropic sphere, rotational symmetry makes the normalized body-geometry operator equal to one third of the identity tensor. No direction inside the sphere is preferred, so each Cartesian axis receives one third of the normalized second-moment contribution. Substitution into the whole-body law gives the exact spherical reduction.

$H_{\text{sphere}} = I/3$
$F_{K,\text{sphere}} = (M \kappa_F/3) [\omega \text{ cross}(G_{\text{phase}} V_{\text{CMB}})]$
$\kappa_F/3 = 1.0246836332340115e-11$

The coefficient $\kappa_F/3$ is not a new universal constant. It is the universal scalar filtered through spherical geometry. The reduction preserves mass extensivity and spin parity. Reversing ω reverses the force. Aligning ω with $G_{\text{phase}} V_{\text{CMB}}$ makes the cross product zero. Perpendicular alignment maximizes the unfiltered cross-product magnitude.

8. Exact Continuous-Rim Reduction

The continuous-rim derivation is the most explicit body-scale closure in the source. Consider a uniform zero-thickness rim of total mass M , radius R , spin axis n , and angular speed ω . A point on the rim has radial unit vector $u(\theta)$, tangent $t(\theta) = n \text{ cross } u(\theta)$, and local velocity $V_E + R \omega t(\theta)$. The elemental force contains a translation-spin term and a rotational self-term.

$r(\theta) = R u(\theta)$
$t(\theta) = n \text{ cross } u(\theta)$
$v(\theta) = V_E + R \omega t(\theta)$
$dF = -\kappa_F dm [\omega B(\theta) + R \omega^2 C(\theta)] u(\theta)$
$B(\theta) = t(\theta)^T G_{\text{phase}} V_E$

$$C(\theta) = t(\theta)^T G_{\text{phase}} t(\theta)$$

8.1 Antipodal cancellation of the self-term

For every rim point at θ there is an antipodal point at $\theta + \pi$. The quadratic self-kernel $C(\theta)$ is even under this shift, while $u(\theta)$ changes sign. Their product is odd and integrates to zero around the complete rim. This exact cancellation removes the radius-dependent $R \omega^2$ self-term from the net force.

$$\int_0^{2\pi} C(\theta) u(\theta) d\theta = 0$$

8.2 Transverse second moment

$$\int u u^T dm = (M/2)(I - n n^T)$$

The remaining translation-spin term is filtered by the rim second moment. The projector $I - n n^T$ selects the plane perpendicular to the spin axis, and the factor $M/2$ is the uniform angular average. The resulting net force is exact for the stated ideal rim model and constitutive order.

$$F_{\text{rim}} = (\kappa_F M \omega/2) n \text{ cross}(G_{\text{phase}} V_E)$$

$$\kappa_F/2 = 1.5370254498510173e-11$$

The net torque about the rim center is zero at this order, and the force lies in the plane perpendicular to $G_{\text{phase}} V_E$. The source force sweep forms a centered ellipse because the allowed axis $n(\alpha)$ is itself constrained to a plane and the cross product maps that axis circle linearly into force space.

Exact continuous-rim force envelope from the stored 0.1-degree sweep

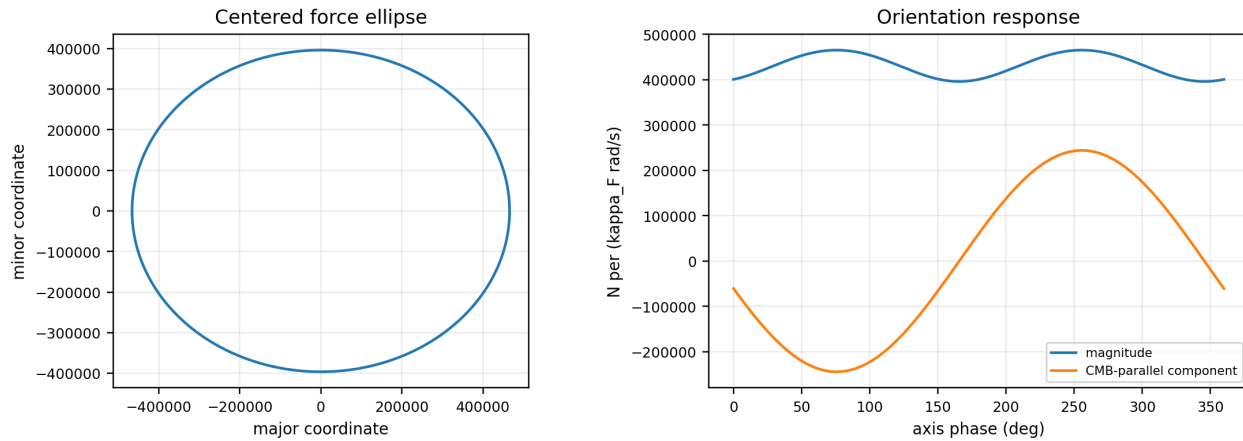


Figure 8. Exact force ellipse and orientation response generated from all 3,601 records in the stored 0.1-degree continuous-rim sweep.

9. Calibrated Engineering Form

Applying the recommended κ_F to the maximum force-envelope magnitude gives the calibrated force slope for a one-kilogram ideal rim. The source stores $1.430702048307761e-5$ N per rad/s, equivalent to $1.4982276814798436e-6$ N per rpm. The inverse slope is 69895.7551072778 rad/s per newton.

Engineering quantity	Value	Unit
Maximum force slope, 1 kg	1.430702048307761e-05	N per rad/s
Inverse slope	69895.7551072778	rad/s per N
Force per rpm	1.498227681479844e-06	N per rpm
Speed for 1 N	6.674552955878311e+05	rpm
Speed for standard 1 kg weight	6.545500474476404e+06	rpm
Standard gravity	9.806649999999999	m/s^2

Calibrated force is linear in mass and angular rate

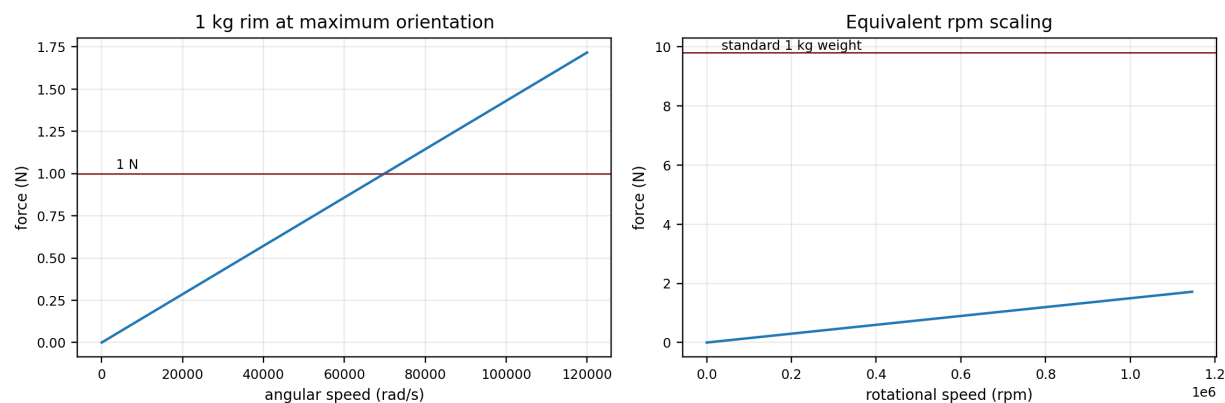


Figure 9. Calibrated linear force scaling with angular speed and rpm for a one-kilogram ideal rim at maximum orientation.

9.1 General scaling laws

Control condition	Exact leading scaling
General	F proportional to $\kappa_F M \omega$
Fixed angular speed	independent of R for an ideal uniform zero-thickness rim
Fixed rim speed v	F proportional to v/R
Fixed angular momentum L	F proportional to L/R^2
Fixed rotational energy E_{rot}	F proportional to $\sqrt{M E_{\text{rot}}}/R$
Spin reversal	$\omega \rightarrow -\omega$ reverses F and preserves the geometric ellipse

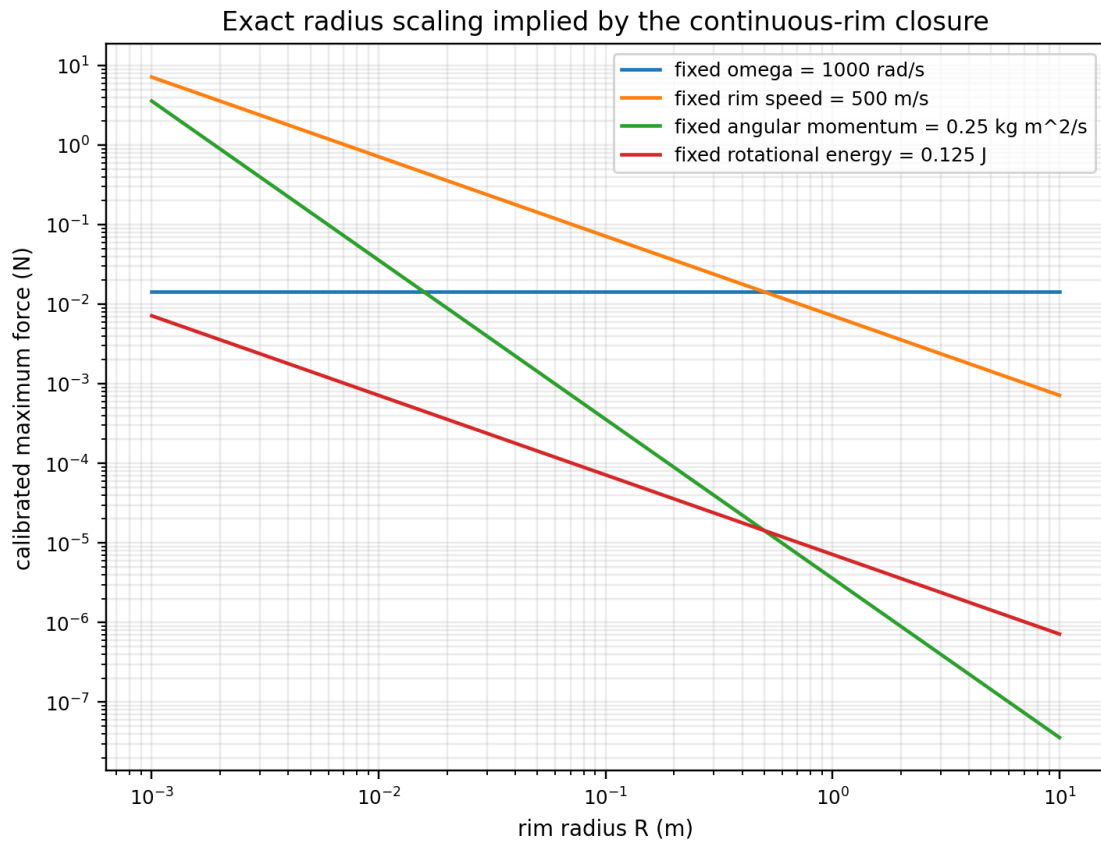


Figure 10. Radius dependence under four distinct control laws. The curves are computed from the exact source scaling relations and the calibrated maximum-orientation slope.

Critical interpretation

The ideal-rim law is scale free only at fixed angular speed. At fixed rim speed, angular momentum, or rotational energy, reducing radius increases angular speed and therefore increases the predicted force. A finite coherence scale or high-frequency correction is not supplied by the present constitutive law.

10. Symmetries, Nulls, and Direction Reversal

10.1 Spin-odd response

$$F(-\omega) = -F(\omega)$$

The force changes sign immediately when the spin direction is reversed while all other vectors remain fixed. This is a direct consequence of linearity in ω . The reversal is not a secondary correction; it is a defining parity of the leading constitutive response.

10.2 Null orientation

$$\omega \text{ parallel } G_{\text{phase}} V_{\text{CMB}} \Rightarrow F_K = 0$$

The null is geometric. It does not depend on the magnitude of the spin or the mass. A large angular speed produces no leading cross-product response when the spin axis is exactly parallel to the transformed translation vector.

10.3 Maximum orientation

ω perpendicular G_{phase} $V_{\text{CMB}} \Rightarrow$ maximum cross-product magnitude

For bodies whose H does not introduce an additional directional suppression, perpendicular alignment maximizes the kernel magnitude. For the constrained continuous-rim sweep, the force magnitude varies between 396230.43448857596 and 465412.6086352174 N per (κ_F rad/s), creating the stored ellipse rather than a circle.

11. Earth-Calibrated Kinematics

The source includes a 366-epoch Earth ledger beginning 2026-07-23. Each record contains barycentric position and velocity, CMB-relative velocity, transformed velocity, common and differential acceleration channels, radial-transverse-normal components, and causal integration states. The ledger allows the coupling to be evaluated over a complete annual cycle rather than at a single orientation.

11.1 Differential acceleration range

Quantity	Value
Earth-Sun differential acceleration range	[5.735920153841129e-10, 7.157363612708431e-10] m/s ²
Maximum tangential acceleration	5.895661182644975e-10 m/s ²
Earth mass	5.972190000000000e+24 kg
Differential force range	[3.425600498356845e+15, 4.274513539418117e+15] N
Unit response magnitude range	[18.6591580328, 23.2831655891] m/s ² per κ_F
Maximum/minimum ratio	1.247814373412331
Half-range fraction about midpoint	0.1102468141246613

Complete 366-epoch Earth response ledger

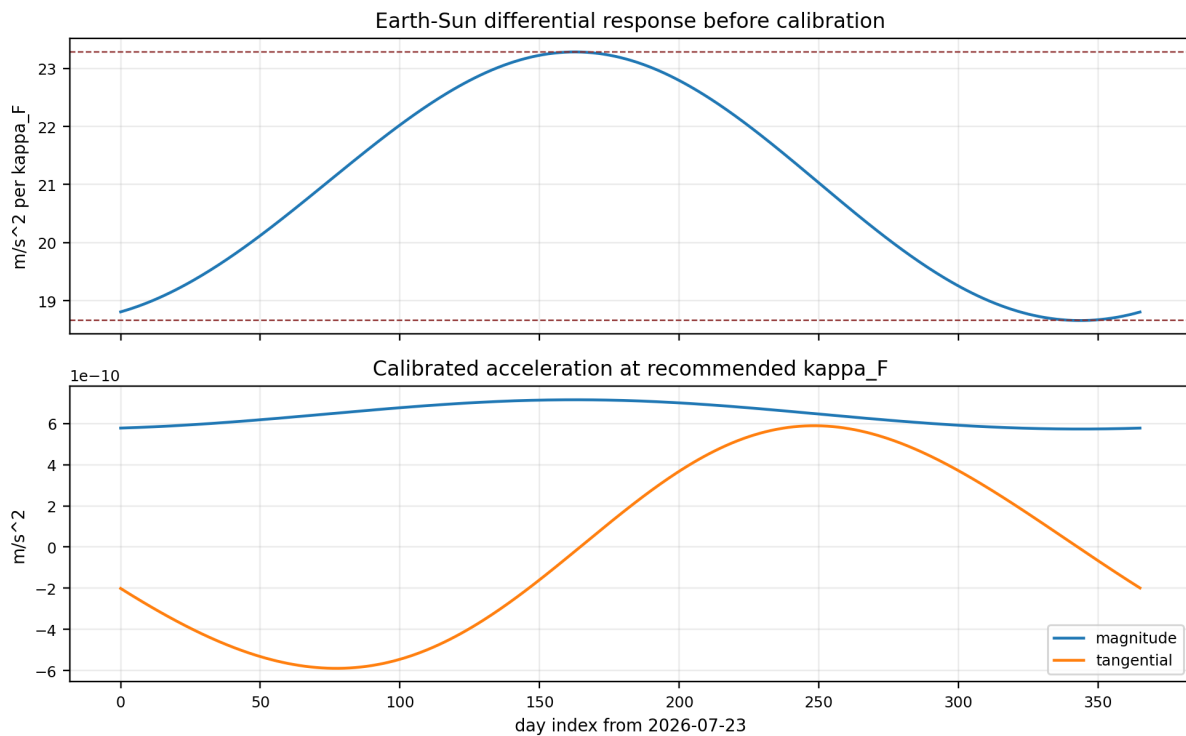


Figure 11. Unit and calibrated Earth-Sun differential response over all 366 stored epochs. No interpolated or synthetic epochs are added.

12. Physical Interpretation of the Coupling

12.1 Weak interface, structured carrier

The property map identifies κ_F as a weak matter-interface coupling with signature 3.0740508997020345e-11. This means that matter responds only weakly per unit of the kinematic kernel. At the same time, G_{phase} is strongly anisotropic and positive definite. The proper interpretation is therefore not a weak field in every sense, but a strongly structured carrier connected to ordinary matter through a small dimensionless susceptibility.

12.2 Susceptibility versus stiffness

A susceptibility tells how strongly matter responds to an applied or pre-existing state. A stiffness tells how much restoring force is generated by a field displacement. The present law contains no field-displacement variable, no restoring-energy function, and no stress-strain relation. Therefore κ_F cannot be converted into an elastic modulus or static stiffness without an additional constitutive equation. This distinction is not semantic; the units and mathematical objects are different.

12.3 High kinematic impedance

The calibrated ideal-rim inverse slope is 69895.7551072778 rad/s per newton for one kilogram at maximum orientation. The source describes this as high kinematic impedance: a large angular rate is required to produce one newton in the reference geometry. Because force scales linearly with coherent mass, increasing mass reduces the required angular rate for a target force in direct inverse proportion, provided the normalized geometry and coherence assumptions remain unchanged.

Derived property	Stored signature	Interpretation
weak matter-interface coupling	3.0740508997020345e-11	suppressed matter response per unit kernel
strong anisotropy	6.156971524705357	principal tensor gain ratio
positive definite	[1.4293287534874834, 2.1125450264474788, 8.800336434665038]	all principal gains positive
transverse rank two	cross-product null parallel to G_{phase} V	orientation-selective response
spin odd	$F(-\omega) = -F(\omega)$	spin-direction sensitivity
mass extensive	F proportional to M	acceleration independent of M for equal normalized geometry
shape dependent	H and second moments	mass distribution enters the response
scale free at leading rim order	no radius dependence at fixed ω	no intrinsic length scale in present rim law
high kinematic impedance	69895.7551072778	large angular rate per unit force

13. Worked Examples

13.1 One-kilogram maximum-orientation rim

For $M=1$ kg and maximum orientation, the calibrated rim law is $F = 1.430702048307761e-5 \omega$ N. At $\omega=1000$ rad/s the force is 0.01430702048307761 N. At 50,000 rpm, $\omega = 5235.987755982989$ rad/s and the predicted force is 0.07491138407399218 N. Reversing the spin changes the sign but not the magnitude.

Input	Value
Mass	1 kg
Angular speed	1000 rad/s
Maximum-orientation force	0.01430702048307761 N
Rotational speed	50,000 rpm

Input	Value
Equivalent angular speed	5235.987755982989 rad/s
Maximum-orientation force at 50,000 rpm	0.07491138407399218 N

13.2 Spherical body with specified kernel

Suppose a one-kilogram spherical body has $|\omega \text{ cross}(G_{\text{phase}} V_{\text{CMB}})| = 1.0\text{e}6 \text{ m/s}^2$. The spherical operator contributes 1/3, so the force magnitude is $1 \text{ kg} \times 3.0740508997020345\text{e-}11 \times (1/3) \times 1.0\text{e}6 \text{ m/s}^2 = 1.0246836332340115\text{e-}5 \text{ N}$. This example isolates the mass-coupling coefficient from the orientation and carrier calculation.

$$F_{\text{sphere}} = M (\kappa_F/3) |\omega \text{ cross}(G_{\text{phase}} V_{\text{CMB}})|$$

13.3 Mass scaling

A ten-kilogram rim with the same normalized geometry, angular rate, and orientation produces ten times the one-kilogram force. At 50,000 rpm and maximum orientation, the ideal leading force is 0.7491138407399218 N. The corresponding acceleration remains 0.07491138407399218 m/s², equal to the one-kilogram acceleration.

14. Measurement and Experimental Pathways

The structure of the law supplies several direct measurement discriminants. These follow from algebraic symmetries rather than from adjustable curve fitting. A measurement program can therefore separate the leading K-Field response from many common mechanical and electromagnetic backgrounds by reversing spin, changing orientation, varying coherent mass, and comparing bodies with different normalized second moments.

14.1 Spin reversal

At fixed laboratory orientation, reversing omega must reverse the leading force vector. A background that is even in spin direction will not follow this sign rule. The comparison should be performed at equal absolute angular speed and matched thermal and vibration states.

14.2 Axis sweep

Sweeping the spin axis through the transformed-vector direction must produce a null, while a perpendicular orientation produces a maximum. For a continuous rim under the stored axis constraint, the complete sweep should trace the calibrated centered ellipse. Both the magnitude and force direction are predicted for every phase.

14.3 Mass and shape sweeps

At equal normalized geometry, force should scale linearly with coherent mass. At equal mass and angular rate, changing the distribution of mass changes H and therefore changes the response. A sphere-to-rim comparison tests the difference between the 1/3 and transverse M/2 geometry factors. A radius sweep at fixed angular speed tests the leading scale-free rim law, while fixed-rim-speed and fixed-energy sweeps test the predicted inverse-radius dependences.

14.4 Annual and diurnal orientation structure

The 366-epoch Earth ledger predicts an annual modulation of the Earth-Sun differential response. A laboratory fixed to Earth also changes orientation through the sidereal day. Combining immediate spin reversal with longer orientation tracking provides a two-timescale signature: a fast odd response controlled by the apparatus and a slow deterministic envelope controlled by Earth motion and orientation.

15. Determinate and Undetermined Properties

The source determines the kinematic susceptibility, carrier anisotropy, positivity, orientation nulls, spin parity, mass extensivity, shape dependence, leading rim scale laws, and calibrated engineering slope. It does not contain the additional state variables and reciprocal equations required to determine several other field properties. Preserving this boundary is necessary for a self-consistent reference.

Property not determined by the present law	Reason additional structure is required
Elastic modulus or static stiffness	requires a field displacement or strain variable and restoring-force law
Energy density	requires a Hamiltonian, Lagrangian, or stored-energy functional
Propagation speed	requires a time-dependent field equation with spatial derivatives
Interaction range	requires a spatial Green function or mediator mass scale
Dispersion	requires frequency- and wave-number-dependent dynamics
Hysteresis or memory	requires internal state variables or a nonlocal temporal kernel
Saturation threshold	requires nonlinear constitutive terms
Backreaction	requires reciprocal momentum exchange with the carrier
Composition dependence	requires material-specific coupling parameters beyond total mass and geometry
Screening or shielding	requires a boundary-value theory for carrier attenuation or redirection
Thermal dependence	requires temperature-dependent constitutive parameters
Quantum response	requires a quantum state space and operator coupling rule

16. Foundational Significance

The mass-coupling law reorganizes rotation, translation, anisotropic geometry, and body shape into one dimensionally closed operator. Instead of treating mass as responding only to a scalar potential or an isotropic background, the framework resolves response through an oriented carrier tensor and a body-dependent second-moment filter. This establishes new pathways for experiment because direction, spin parity, mass, shape, radius control law, and Earth phase are all independently testable coordinates of the same equation.

The small scalar κ_F provides a quantitative bridge between the carrier-level kinematic kernel and matter-level acceleration. The Cell3 geometry supplies the anisotropic foundation, G_{phase} supplies the principal directional gains, H supplies the coherent body filter, and the cross product supplies the transverse time-odd response. Together these objects form a compact explanatory structure with fewer independent assumptions than a collection of unrelated orientation effects.

Archival statement

The full K-Field mass-coupling factor is the operator $M \kappa_F H$, with $\kappa_F = 3.0740508997020345e-11$. The scalar is dimensionless, the operator is mass extensive and shape dependent, the active kernel is $\omega_{\text{cross}}(G_{\text{phase}} V_{\text{CMB}})$, the sphere reduction is $\kappa_F/3$, and the ideal-rim reduction is $\kappa_F/2$.

Appendix A. Complete 27-Node Cell3 Ledger

The following table lists every centered half-step node used in the figures. Integer address $L=(i,j,k)$ maps to ICRS native coordinate $p=(1/2)B L$. Node type is determined by the number of zero address components. The table is generated directly from the source node arrays.

Address	Type	ICRS x	ICRS y	ICRS z	lambda=i+j+k
(-1,-1,-1)	corner	-0.010974902499961	0.0073859831274229	0.039661911011939	-3
(-1,-1,0)	edge center	-0.011290092242635	-0.004742877881141	0.018199733726392	-2
(-1,-1,1)	corner	-0.011605281985309	-0.016871738889705	-0.0032624435591546	-1
(-1,0,-1)	edge center	-0.0098755295157995	0.013837068256769	0.022628660243911	-2
(-1,0,0)	face center	-0.010190719258473	0.0017082072482053	0.0011664829583645	-1
(-1,0,1)	edge center	-0.010505909001147	-0.010420653760359	-0.020295694327182	0
(-1,1,-1)	corner	-0.0087761565316379	0.020288153386115	0.0055954094758835	-1
(-1,1,0)	edge center	-0.0090913462743115	0.0081592923775515	-0.015866767809663	0
(-1,1,1)	corner	-0.0094065360169852	-0.0039695686310124	-0.03732894509521	1
(0,-1,-1)	edge center	-0.00078418324148799	0.0056777758792177	0.038495428053575	-2
(0,-1,0)	face center	-0.0010993729841617	-0.0064510851293463	0.017033250768028	-1
(0,-1,1)	edge center	-0.0014145627268354	-0.01857994613791	-0.0044289265175191	0
(0,0,-1)	face center	0.00031518974267369	0.012128861008564	0.021462177285547	-1
(0,0,0)	body center	0	0	0	0
(0,0,1)	face center	-0.00031518974267369	-0.012128861008564	-0.021462177285547	1
(0,1,-1)	edge center	0.0014145627268354	0.01857994613791	0.0044289265175191	0
(0,1,0)	face center	0.0010993729841617	0.0064510851293463	-0.017033250768028	1
(0,1,1)	edge center	0.00078418324148799	-0.0056777758792177	-0.038495428053575	2
(1,-1,-1)	corner	0.0094065360169852	0.0039695686310124	0.03732894509521	-1
(1,-1,0)	edge center	0.0090913462743115	-0.0081592923775515	0.015866767809663	0
(1,-1,1)	corner	0.0087761565316379	-0.020288153386115	-0.0055954094758835	1
(1,0,-1)	edge center	0.010505909001147	0.010420653760359	0.020295694327182	0
(1,0,0)	face center	0.010190719258473	-0.0017082072482053	-0.0011664829583645	1
(1,0,1)	edge center	0.0098755295157995	-0.013837068256769	-0.022628660243911	2
(1,1,-1)	corner	0.011605281985309	0.016871738889705	0.0032624435591546	1
(1,1,0)	edge center	0.011290092242635	0.004742877881141	-0.018199733726392	2
(1,1,1)	corner	0.010974902499961	-0.0073859831274229	-0.039661911011939	3

Appendix B. Complete Boundary-Plane Equations

Family	Member	n_x	n_y	n_z	d	Equation form
ab faces (constant c coefficient)	1	0.19409220080777	0.91317716882123	0.3583792347908	-0.018828573500193	$0.1940922 x + 0.91317717 y + 0.35837923 z + -0.018828574 = 0$
ab faces (constant c coefficient)	2	0.19409220080777	0.91317716882123	0.3583792347908	0.018828573500193	$0.1940922 x + 0.91317717 y + 0.35837923 z + 0.018828574 = 0$
ac faces (constant b coefficient)	1	0.089052594101605	0.86657676929103	-0.49103394832628	0.014052166903391	$0.089052594 x + 0.86657677 y + -0.49103395 z + 0.014052167 = 0$
ac faces (constant b coefficient)	2	0.089052594101605	0.86657676929103	-0.49103394832628	-0.014052166903391	$0.089052594 x + 0.86657677 y + -0.49103395 z + -0.014052167 = 0$
bc faces (constant a coefficient)	1	-0.99596519030896	0.083602162507377	-0.0326192905644	-0.010254361619054	$-0.99596519 x + 0.083602163 y + -0.032619291 z + -0.010254362 = 0$
bc faces (constant a coefficient)	2	-0.99596519030896	0.083602162507377	-0.0326192905644	0.010254361619054	$-0.99596519 x + 0.083602163 y + -0.032619291 z + 0.010254362 = 0$

Appendix B.1 Face areas and normals

Face	Area	Included angle	Unit normal ICRS
A_ab	0.000754721789672624	83.9334830123127 deg	(0.194092200808, 0.913177168821, 0.358379234791)
A_ac	0.00101125575768807	80.4481300183378 deg	(0.0890525941016, 0.866576769291, -0.491033948326)
A_bc	0.00138578443173329	50.3632318137562 deg	(-0.995965190309, 0.0836021625074, -0.0326192905644)

Appendix C. Complete Matrix Record

Canonical basis columns [a b c]

row	1	2	3
1	0.02079705878161049	0.003856816104884186	0.008182284351835403
2	0	0.03628983297903845	0.03076238244505114
3	0	0	0.03765714700071637

ICRS basis columns [a b c]

row	1	2	3
1	0.02038143851694645	0.002198745968323355	-0.000630379485347372
2	-0.003416414496410549	0.01290217025869254	-0.02425772201712787
3	-0.002332965916728961	-0.03406650153605571	-0.04292435457109382

Canonical-to-ICRS rotation

row	1	2	3
1	0.9800154305938903	-0.04356573722285733	-0.1940922008128713
2	-0.164273926053018	0.3729899939226731	-0.9131771688158449
3	-0.1121776853750038	-0.9268120041202775	-0.3583792347772863

Direct metric M

row	1	2	3
1	0.0004325176539657619	8.021043124313843e-05	0.0001701674486329726
2	8.021043124313843e-05	0.001331827008113401	0.001147919287031111
3	0.0001701674486329726	0.001147919287031111	0.002431334671143443

Reciprocal metric R

row	1	2	3
1	2377.51218408627	0.3962859857098399	-166.5875502838364
2	0.3962859857098177	1266.057440195837	-597.7783339805006
3	-166.5875502838364	-597.7783339805006	705.1884208705073

Carrier tensor G_phase

row/column	x	y	z
x	2.9764106378	1.2431507795	-2.0803600354
y	1.2431507795	2.5922286635	-2.3178839937
z	-2.0803600354	-2.3178839937	6.7735709133

Carrier eigenvectors as columns

row	mode 1	mode 2	mode 3
1	-0.4089530672843345	-0.8285538959646047	-0.3824340861385266
2	0.8845250094313242	-0.2568323054783916	-0.3894260835552364
3	0.2244390706834493	-0.4975295049989798	0.8379090017455477

Appendix D. Symbol Dictionary

Symbol	Meaning	Units
B	Cell3 primitive basis matrix with columns a,b,c	native length
$L=(i,j,k)$	integer Cell3 node address	dimensionless
$p(L)$	ICRS node position $(1/2)BL$	native length
M_metric	direct Cell3 metric $B^T B$	native length ²
R_metric	reciprocal metric M_metric^{-1}	native length ⁻²
G_phase	symmetric anisotropic carrier tensor	dimensionless
V_CMB or V_E	declared translation velocity	m/s
g	transformed translation $G_phase V_CMB$	m/s
ω	angular velocity vector	s ⁻¹
H	normalized body-geometry operator	dimensionless
κ_F	universal matter-interface susceptibility	dimensionless
C_M	complete mass-coupling operator $M \kappa_F H$	kg
a_K	K-Field whole-body acceleration	m/s ²
F_K	K-Field whole-body force	N
n	rim spin-axis unit vector	dimensionless
$u(\theta)$	rim radial unit vector	dimensionless
$t(\theta)$	rim tangent $n \times u(\theta)$	dimensionless

Appendix E. Source Artifacts, Reproducibility, and Machine Record

Filename	Size (bytes)	SHA-256
K-Field_mass_harmonics_1784575932.json	28761	a8a90ad5ad3c049c660af65eb245414b730f87d3217db12ae2bc24472a370210
K-Field_earth_kinematic_derivation_of_coupling_constant_1784828045.json	978067	2f9d605451b26f29a84ff7511dd69efa3d53db6212a586b0b6562dc54faf1477
K-Field_flywheel_force_analysis_1784808116.json	2316139	a965becf29845009e05ca992befb7699a5869415b5296c315bdf81fef10fa4e8
K-Field_flywheel_force_distribution_disk_1784810003.json	31792574	6501035abae5ed5ff29aaa60906be4dc03ac639d0834abb74a3474e2dee06156

The companion machine-readable file is `K-Field_mass_coupling_factor_1785088666.json`. It contains the exact scalar and operator definitions, calibration channels, complete Cell3 geometry used in this report, carrier data, rim equations and envelope, Earth calibration, all 366 plotted ledger records, figure hashes, report hash, and generation metadata. The report figures can therefore be regenerated without digitizing the PDF.

Generated figure hashes

Figure file	SHA-256
F01_title_composite.png	f36ce6ae9723a0e0de4f49bf8db10a068e15202e204acaef61326946ffd977ae
F02_cell3_all_nodes_planes.png	c80bb80051a699d97c3e7ab99cb1c5ffdb94b9c72e0577d2ddf19bb15f83dc4b
F03_six_boundary_planes_all_nodes.png	81bd07c262561a896f0a184f34f62585fa10630be1150bc545d7906b434fcd1e
F04_metrics_and_eigenvalues.png	5f307ea4582dcf6b8493180c32d3f59c5f1aa7cfa11fbd330a55fcd9d1a216f3
F05_carrier_vector_geometry.png	d4e1ba3949da2ab9e1f560866e7f974b04fa9b8cd904fafa64c7ceda478e172e
F06_calibration_channels.png	527d48cb07df6e2b1a73e8894148f6255e204641c7bda9f864653ddc853655e5
F07_force_envelope.png	5f3083fb861d7b1a464eed822e634efeeae87d1075753929237c152d14e9d1a88
F08_engineering_scaling.png	a877c1297d12ef3af8bfa82400a0331900c3af98d00259250fc04cd7f40136c4
F09_radius_scaling.png	cf2cebbafaa713898f6fa80316c69b06c52b84198cb3175c627b6aa18624ae38
F10_earth_366_epoch_modulation.png	a7da0091c65ef94c1e3832461090313968a758b4748a6f407f43b2cf0840a18b
F11_mass_extensivity.png	c0451c4101ff6cb38454ec0adfc4626c218fbb32f6cb84a554d13902453f94be

End of archival reference

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